Mid Semester Examination 2025

Name of the Program:	B. Tech (Civil Engineering),	*Semester*	: III
Name of the Course	: Fluid Mechanics	*Course Code*	: TCE 301
Time	: 1-1/2 Hour	**Max. Marks**	: 50

Note:

i) Answer **all the questions** by choosing ***any one of the sub questions***.

ii) Each question carries 10 marks

Q 1. 10 Marks

a) With the help of a neat diagram define the term; Absolute pressure, Gauge pressure and vacuum pressure. What is the relation between Absolute pressure and Gauge pressure? **CO 1**

(OR)

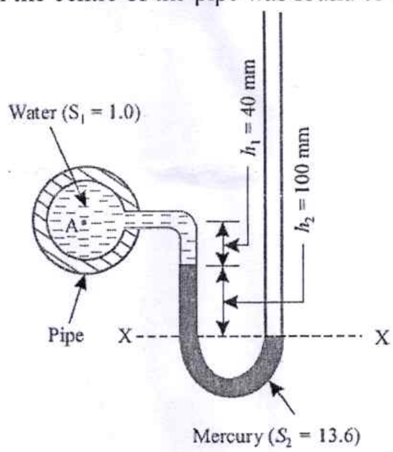
b) Calculate the capillary effect in millimeters in a glass tube of diameter 4 mm, when immersed in (i) water and (ii) mercury. The values of surface tension of water and mercury are 0.073575 N/m and 0.51N/m respectively. The angle of contact for water is zero and that for mercury is 130^0. **CO 1**

Q 2. 10 Marks

a) What are various devices adopted for measuring fluid pressure in different conditions? Illustrate them with the help of neat diagram. **CO 1**

(OR)

b) U-tube manometer containing mercury was used to find the negative pressure in the pipe, containing water. The right limb was open to the atmosphere. Find the vacuum pressure in the pipe, if the difference of mercury level in the two limbs was 100 mm and height of water in the left limb from the centre of the pipe was found to be 40 mm below. **CO 1**

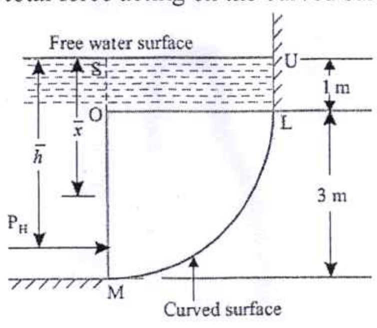


Q 3. 10 Marks

a) Define Metacentre and Metacentric height. How are they important in case of floating body? **CO 2**

(OR)

b) Figure shows a curved surface LM, which is in the form of a quadrant of a circle of radius 3 m, immersed in the water. If the width of the gate is unity, calculate the horizontal and vertical components of the total force acting on the curved surface. **CO 1**



Q 4. 10 Marks

a) What do you mean by flow net? Explain the method of drawing flow net. **CO 2**

(OR)

b) If the velocity component for two dimensional flow are given by $u = \frac{x}{x^2 + y^2}$ and $v = \frac{y}{x^2 + y^2}$; determine the acceleration components a_x and a_y. **CO 2**

Q 5. 10 Marks

a) Explain the terms: Path Line, Stream Line, Streak Line and Stream Tube. **CO 2**

(OR)

b) The diameters of a pipe at the sections 1-1and 2-2 are 200 mm and 300 mm respectively. If the velocity of water flowing through the pipe at section 1-1 is 4m/s, find: **CO 2**

(i) Discharge through the pipe, and

(ii) Velocity of water at section 2-2